

## EXP 3 : Classification with Decision Trees

### Aim

To implement a **Decision Tree classifier** for the Pet Adoption dataset and evaluate its performance using **accuracy score and confusion matrix**.

### Algorithm

1. Import the required libraries.
2. Load the Pet Adoption dataset.
3. Handle missing values.
4. Encode categorical variables.
5. Separate the features and target variable.
6. Split the dataset into training and testing sets.
7. Initialize the DecisionTreeClassifier.
8. Train the Decision Tree model using the training data.
9. Predict the test data.
10. Calculate the confusion matrix and accuracy score.
11. Visualize the confusion matrix.
12. Visualize the Decision Tree.

### Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix
print("Libraries imported successfully!")
df = pd.read_csv("pet_adoption.csv")
print("\nDataset loaded successfully!")
print("Dataset Shape:", df.shape)
display(df.head())
print("\nMissing Values Before Preprocessing:")
print(df.isnull().sum())
df["Breed"] = df["Breed"].fillna(df["Breed"].mode()[0])
```

```

df["Vaccinated"] = df["Vaccinated"].fillna(df["Vaccinated"].mode()[0])
df["Fee"] = df["Fee"].fillna(df["Fee"].median())
print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())
X = df.drop("AdoptionSpeed", axis=1)
y = df["AdoptionSpeed"]
print("\nFeatures Shape:", X.shape)
print("Target Shape:", y.shape)
categorical_cols = X.select_dtypes(
    include=["str", "object"]
).columns
X_encoded = X.copy()
label_encoders = {}
for col in categorical_cols:
    le = LabelEncoder()
    X_encoded[col] = le.fit_transform(
        X_encoded[col].astype(str)
    )
    label_encoders[col] = le
print("\nCategorical columns encoded:")
print(list(categorical_cols))

X_train, X_test, y_train, y_test = train_test_split(
    X_encoded,
    y,
    test_size=0.30,
    random_state=42,
    stratify=y
)
print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
dt_model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=5,
    random_state=42
)
dt_model.fit(X_train, y_train)
print("\nDecision Tree model trained successfully!")
y_pred = dt_model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

```

```

print("\n=====")
print("DECISION TREE RESULTS")
print("=====")
print("Accuracy Score:", round(accuracy, 4))
print("Accuracy Percentage:", round(accuracy * 100, 2), "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
plt.figure(figsize=(7, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    cbar=False
)
plt.title("Decision Tree - Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.tight_layout()
plt.show()
plt.figure(figsize=(24, 12), dpi=150)
plot_tree(
    dt_model,
    feature_names=X_encoded.columns,
    class_names=[str(x) for x in sorted(y.unique())],
    filled=True,
    rounded=True,
    max_depth=3,
    fontsize=10
)
plt.title(
    "Decision Tree for Pet Adoption Prediction",
    fontsize=20
)
plt.tight_layout()
plt.show()
print("Dataset Size:", df.shape)
print("Training Samples:", len(X_train))
print("Testing Samples:", len(X_test))

```

```
print("Number of Classes:", y.nunique())
print("Decision Tree Accuracy:",
      round(accuracy * 100, 2), "%")
```

## Output:

Libraries imported successfully!

Dataset loaded successfully!

Dataset Shape: (500, 14)

| PetType | AgeMonths | Gender | Breed       | Vaccinated | Dewormed | Sterilized | Health  | MaturitySize | FurLength | Fee    | PhotoAmount | VideoAmount | AdoptionSpeed |
|---------|-----------|--------|-------------|------------|----------|------------|---------|--------------|-----------|--------|-------------|-------------|---------------|
| Dog     | 64        | Male   | Beagle      | Yes        | No       | Yes        | Healthy | Small        | Medium    | 732.0  | 4           | 0           | 1             |
| Cat     | 86        | Male   | Persian     | No         | Yes      | No         | Healthy | Large        | Medium    | 1748.0 | 6           | 0           | 2             |
| Dog     | 33        | Female | Mixed Breed | No         | Yes      | Yes        | Healthy | Medium       | Medium    | 3924.0 | 9           | 2           | 3             |
| Dog     | 88        | Male   | Beagle      | No         | Yes      | Yes        | Healthy | Small        | Medium    | 423.0  | 10          | 0           | 3             |
| Dog     | 34        | Male   | Mixed Breed | Yes        | Yes      | Yes        | Healthy | Large        | Short     | 3478.0 | 10          | 1           | 2             |

Missing Values Before Preprocessing:

```
PetType      0
AgeMonths    0
Gender       0
Breed        5
Vaccinated   5
Dewormed     0
Sterilized   0
Health       0
MaturitySize 0
FurLength    0
Fee          5
PhotoAmount  0
VideoAmount  0
AdoptionSpeed 0
dtype: int64
```

Missing Values After Preprocessing:

```
PetType      0
AgeMonths    0
Gender       0
Breed        0
Vaccinated   0
Dewormed     0
Sterilized   0
Health       0
MaturitySize 0
FurLength    0
Fee          0
PhotoAmount  0
VideoAmount  0
AdoptionSpeed 0
dtype: int64
```

Features Shape: (500, 13)

Target Shape: (500,)

Categorical columns encoded:

```
['PetType', 'Gender', 'Breed', 'Vaccinated', 'Dewormed', 'Sterilized', 'Health', 'MaturitySize', 'FurLength']
```

Training Data Shape: (350, 13)

Testing Data Shape: (150, 13)

Decision Tree model trained successfully!

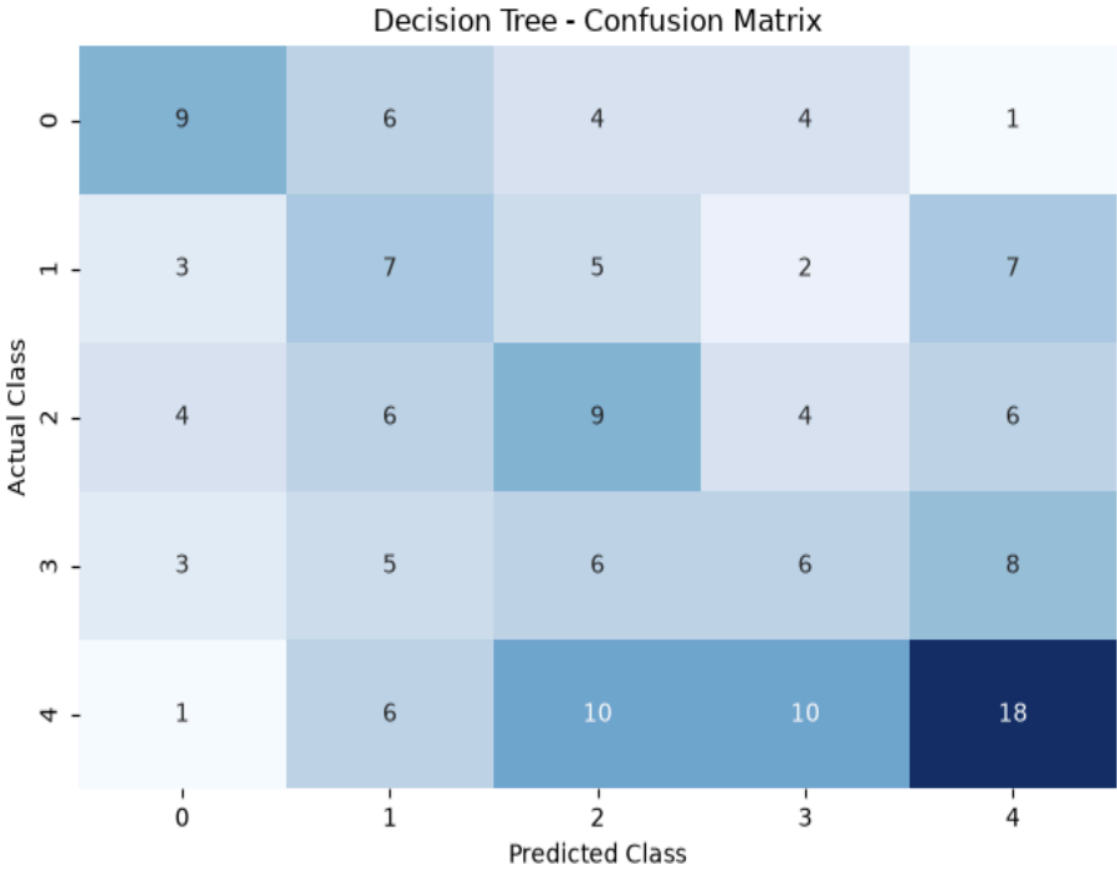
=====

DECISION TREE RESULTS

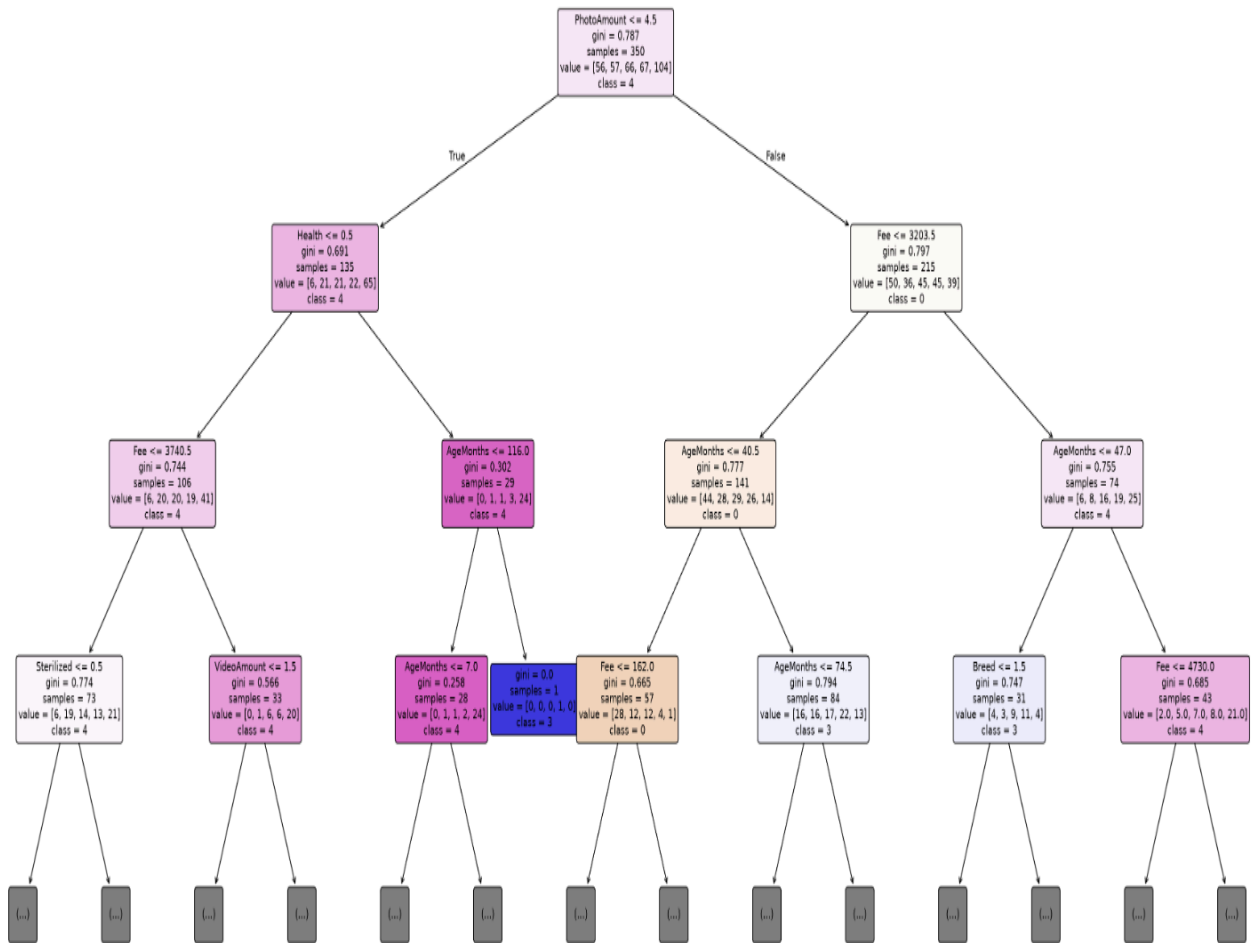
=====

Accuracy Score: 0.3267  
Accuracy Percentage: 32.67 %

Confusion Matrix:  
[[ 9 6 4 4 1]  
 [ 3 7 5 2 7]  
 [ 4 6 9 4 6]  
 [ 3 5 6 6 8]  
 [ 1 6 10 10 18]]



Decision Tree for Pet Adoption Prediction



Dataset Size: (500, 14)

Training Samples: 350

Testing Samples: 150

Number of Classes: 5

Decision Tree Accuracy: 32.67 %

## Result:

The Decision Tree classification model was successfully implemented on the Pet Adoption dataset. The model was trained and tested, and its performance was evaluated using the **accuracy score and confusion matrix**. The Decision Tree was also visualized to show the classification process.